COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-DBS-005: Database Backup and Recovery

Document ID	TS-DBS-005
Title	Database Backup and Recovery
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	4.0
Effective Date	01/28/2021
Last Updated	07/01/2026

Purpose

This standard is issued under the authority of the Governor's Office of Information Technology (OIT) (C.R.S. 24-37.5, et seq.) and the Office of Enterprise Architecture (Rule 8 CCR 1501-7). It communicates the standards OIT has developed for database backup and recovery in all database environments on DBS-supported database platforms.

Standards are based on a best-of-value decision assessment involving multiple factors, such as stakeholder needs, hardware configurations serviced and supported by OIT, and operational and security best practices.

Scope

These standards apply to OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive. This standard shall in no way require OIT to support any products and services not approved by OIT.

For the purpose of enforcement, this standard is rated CAT 1 according to the Enforcement Category Chart.

Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

DBS Team

- Performs all database-related installations, migrations, upgrades, patching, and administration of databases on the database server.
- Ensures data and database structures are backed up, stored on secure storage, accessible only to authorized personnel, and can be recovered when required.
- Ensures production database backups are kept separate from non-production database backups.
- Manages database user security on supported database platforms.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

Standards

- DBS will be responsible for the backup and recovery of all DBS-supported databases.
- Database backup, archival, and retention will follow federal statutes, state statutes, and business requirements.
- In the absence of any of the above retention policies, DBS will retain backups for a maximum of 30 days.
- Additional levels of database backup, retention, and recovery, as requested by the data owner, will be reviewed by DBS.

Variance and Exception Process

Requests for variances from any provision of this standard must be requested using the Technical Standard End User Exception Request Procedure.

The Exception Request process addresses business or technical reasons for the inability to adhere to one or more requirements in this standard. Requests for exceptions should include compensating controls — processes, technologies, or procedures proposed to mitigate any deviation from the standard requirement. Use the Secure Configuration Exception Request

form; see the How-to Guide for Secure Configuration Exception Requests or Policy Exception Requests in ServiceHub. Exception requests will be processed and recommended for approval, conditional approval, or denial.

References

- OIT documents and policies
- State of Colorado Style Library
- TS-OEA-001: Technology Accessibility for Persons with Disabilities
- Enforcement Category Chart
- State documents and policies
- Colorado Brand Guidelines
- C.R.S. 24-37.5-101, et seq.
- CISP-001: Access Control
- TS-CISO-001: Data Security Categorization
- TS-INF-003: Identity and Access Administration

Definitions

- Standard: A rule that specifies a particular course of action or response to a given situation. Standards are mandatory directives to carry out management's policies and are used to measure compliance with policies.
- DBS: Database Services
- DBA: Database Administrator
- System Administrator or SysAdmin
- CIS: Center for Internet Security
- CISO: Chief Information Security Officer
- OEA (Office of Enterprise Architecture): Formed as an office under OIT by Governor's Rule 8 CCR 1501-7.
- OIT (Office of Information Technology): Created and governed by C.R.S. 24-37.5, et seq.
- Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception, subject to review and approval by an authority in the area.